Part 2 technical learning report

SPS: lineage, equations, and algorithm

Checked predecessors, correction, and claim limits

Result in one sentence

SPS adapts the path-space sampler family to lattice field theory and uses the full trajectory ratio as an extended-space independence-MH correction.

Question	Answer from the checked sources
Did SPS introduce path-space sampling?	No. PIS and DDS already train stochastic paths for unnormalized targets.
Did SPS first learn both directions?	No. CMCD already adapts coupled forward and backward drifts.
What is different here?	Independent drift networks, learned scalar diffusion, a lattice implementation, and extended-space trajectory IMH.
What is not established?	First-ever priority, isolated component benefit, support robustness, or matched total-cost speedup.

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T0–T3 verified; T4–T5 not requested

Source IDs and paper anchors are retained in the companion Markdown and CSV files.

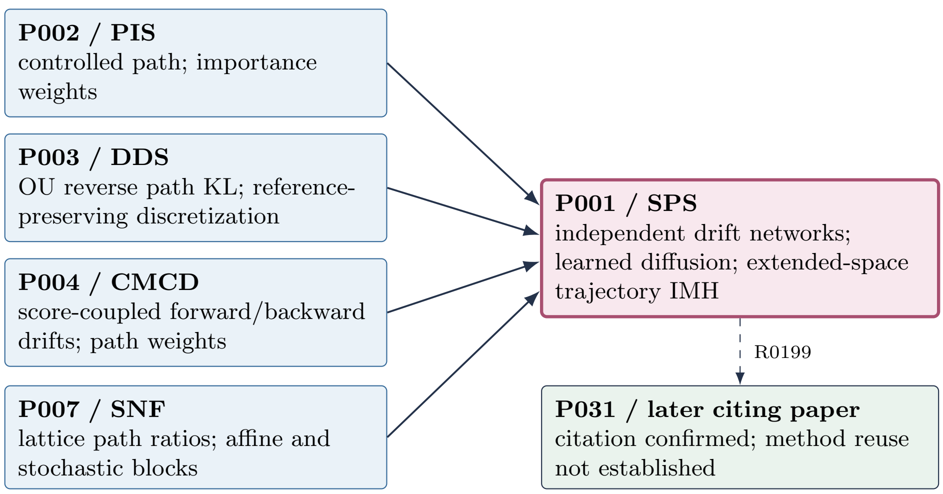
1 Question and reading contract

The task was not to summarize SPS in isolation. It was to determine which parts were already present in PIS, DDS, CMCD, and stochastic normalizing flows, then reconstruct the corrected SPS algorithm without overstating novelty. Five exact paper versions were read at C4 level and passed the native R0–R6 record validator.

ID	Paper	Exact version	Role in the subtraction
P002	Path Integral Sampler	2111.15141v2	controlled path and trajectory weights
P003	Denoising Diffusion Samplers	2302.13834v2	OU reverse path KL
P004	Controlled Monte Carlo Diffusions	2307.01050v12	paired drift adaptation
P007	Stochastic normalizing flows	2201.08862v3	lattice path ratios and reweighting
P001	Stochastic Path Sampler	2606.13790v1	focal lattice sampler and trajectory IMH

P004 was added during reading because P001 states that CMCD already learns forward and backward drifts. It is therefore required for the novelty check.

2 Checked lineage



Direct-citation relations: P002/R0171, P003/R0177, P004/R0178, and P007/R0172. R0199 confirms a later citation only. Dijkstra set the reading order; paper sections and equations support the claims.

3 Predecessor subtraction

Dimension	Already present	SPS form	Decision
Path sampling	PIS and DDS learn paths for unnormalized targets	Discrete lattice Langevin path	inherited family
Path objective	PIS, DDS, CMCD, and SNF use path costs or path ratios	$D_{KL}(q_F \| \widetilde{q}_B)$	formulation differs
Both directions	CMCD adapts score-coupled forward/backward drifts	independent drift networks plus learned $\sigma_\theta(t)$	difference confirmed; benefit open
Lattice path branch	SNF combines stochastic updates and learned affine maps	learned stochastic drift kernels without invertible affine blocks	architecture changed
Correction	predecessors use trajectory weights or reweighting	full-trajectory extended-space IMH	narrowest checked difference
Efficiency	papers report incompatible ESS, error, or correlation metrics	SPS reports chain autocorrelation and wall times	no speed ranking

The current sources support a design difference. They do not support a *first-ever* claim. Establishing priority would require a separate high-recall search for earlier auxiliary-variable and path-space MH constructions.

4 How the corrected SPS algorithm works

1. Draw $s_0 \sim \pi_0$ from the tractable Gaussian prior.
2. Simulate a complete forward path with $K_{\theta,F}$ and $\sigma_\theta(t)$; retain every Gaussian transition factor.
3. Evaluate $K_{\theta,B}$ and the backward transition factors on the same path.
4. Train both drifts and the diffusion network by minimizing the sampled forward/unnormalized-backward path log-ratio.
5. Generate an independent new path and retain its path-dependent proposal ratio.
6. Compare the proposed and current extended paths with Eq. (2.19); accept the proposed endpoint or retain the current one.
7. Measure observables and autocorrelation on the corrected endpoint chain.

$$s_{i+1} = s_i + \sigma_\theta^2(t_i) K_{\theta,F}(s_i, t_i)\Delta t + \sigma_\theta(t_i)\xi_i\sqrt{\Delta t}, \qquad \text{P001 Eq. (2.2)}, \tag{1}$$

$$D_{KL} = \int dq_F(\tau) \log \frac{dq_F(\tau)}{d\widetilde{q}_B(\tau)}, \qquad \text{P001 Eq. (2.13)}, \tag{2}$$

$$P_{\text{acc}} = \min\left[1, \frac{\widetilde{\pi}(s'_T) T(s'_T \to s_T)}{\widetilde{\pi}(s_T) T(s_T \to s'_T)}\right], \qquad \text{P001 Eqs. (2.18)–(2.19)}. \tag{3}$$

The first equation generates a path, the second trains its forward/backward balance without knowing Z, and the third restores target invariance if the proposal covers the target support. Evidence: `E-260613790-M` and `E-260613790-L`.

5 Correction mechanism

Paper	Finite-training output	Trajectory information	Correction object
PIS	weighted endpoints	dQ^*/dQ^u	importance estimator
DDS	approximate endpoints plus $\widehat{Z}$	dP^{ref}/dQ_θ	importance estimator
CMCD	approximate endpoints plus weight	forward/backward path ratio	importance estimator
SNF	weighted lattice paths	reverse/forward transition ratio	reweighted observable
SPS	correlated corrected chain	path-dependent proposal ratio	extended-space IMH

All five methods use full-path information but use it differently. This comparison prevents an incorrect claim that SPS introduced trajectory correction.

6 Evidence and claim limits

Claim	Source-supported result	Boundary
Corrected observables	SPS+IMH follows the reported HMC magnetization and susceptibility across the tested 2D $L \times 8$ scan (`E-260613790-R`)	benchmark-specific; uncorrected tails and multimodal regions deviate
Acceptance	about 0.60–0.76 at $L = 16$ and 0.45–0.62 at $L = 64$ (P001 Fig. 6)	acceptance is not total efficiency
Timing	about 1.1–1.2 h training and 1.1–2.1 min generation across the reported sizes (P001 Table 4)	no matched HMC GPU-hour ratio (`E-260613790-L`)
Autocorrelation	about 0.5 for SPS+IMH and about 160 for HMC at the reported $L = 64$, $\kappa = 0.27$ point (P001 Fig. 8)	IMH steps and HMC trajectories are different units; not a speed ratio
Exactness	extended-space IMH targets the physical distribution (`E-260613790-M`)	proposal support is required (`E-260613790-L`)

Allowed: In the tested two-dimensional benchmark, corrected SPS reproduces the reported HMC observables; the paper leaves a matched total-cost comparison open.

Do not write: SPS is universally 320 times faster than HMC.

7 Validation and run accounting

Level	Demonstration	Status
T0	exact identities, versions, URLs, pages, and hashes	**confirmed**
T1	bounded explanation of problem, mechanism, result, and limits	**confirmed**
T2	six aligned formula roles and five validated reading records	**confirmed**
T3	ordered training, proposal, extended-state correction, and output	**confirmed**
T4	equation-to-code symbol mapping	not requested
T5	numerical reproduction	not requested

Goal-mode telemetry is recorded in `goal_usage_snapshots.csv`. The counter is the Goal runtime's cumulative `tokensUsed`, not API billing usage or monetary cost.

Stage	Work	Cumulative tokens	Cumulative seconds
S00	goal start	0	0
S01	contract and Part 1 lock	14,413	39
S02	exact source lock	395,432	3,945
S03	five equation-level readings	711,191	5,130
S04	predecessor subtraction and review	763,038	5,396
S05	outputs and visual layout check	915,839	5,817
S06	final pre-completion audit	1,083,239	6,220
S07	Goal completion	1,161,662	6,367

8 Companion audit files

Artifact	What it records
`paper_reading_record_P*.md`	exact paper-level extraction and claim boundaries
`paper_reading_ledger.csv`	one-row comparison across the five core papers
`innovation_delta.csv`	inherited, changed, inferred, and unresolved components
`equation_code_map.csv`	formula-to-algorithm trace; T4 fields explicitly not requested
`lineage_learning_path.mmd`	checked citation route and unresolved successor edge
`review_core.md`	retained, weakened, rejected, and open claims
`goal_usage_snapshots.csv`	stage-by-stage Goal token and elapsed-time observations

9 Source index

- P001: arXiv:2606.13790v1, `E-260613790-I`, `E-260613790-M`, `E-260613790-R`, `E-260613790-L`.
- P002: arXiv:2111.15141v2, `E-211115141-I`, `E-211115141-M`, `E-211115141-L`.
- P003: arXiv:2302.13834v2, `E-230213834-I`, `E-230213834-M`, `E-230213834-L`.
- P004: arXiv:2307.01050v12, `E-230701050-I`, `E-230701050-M`, `E-230701050-L`.
- P007: arXiv:2201.08862v3, `E-220108862-I`, `E-220108862-M`, `E-220108862-L`.